

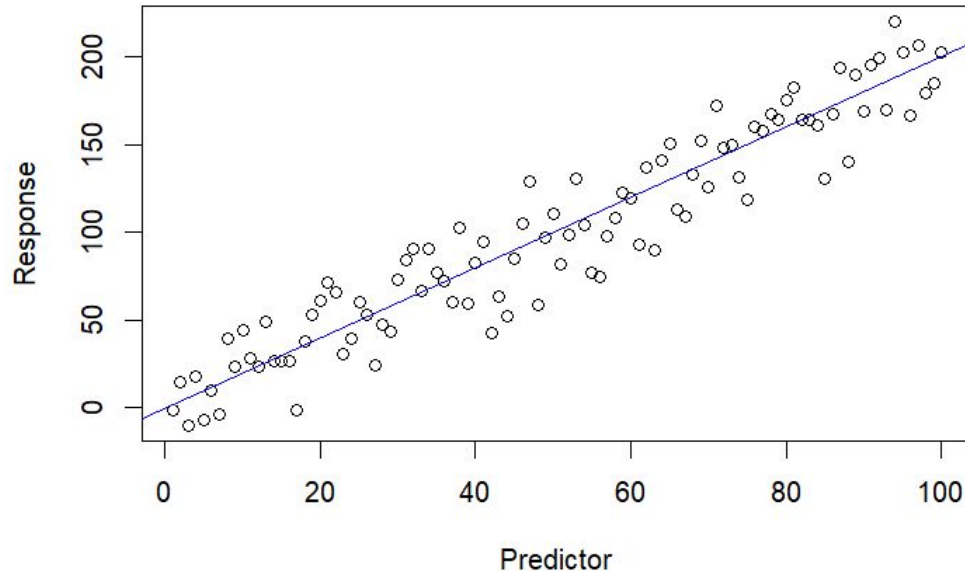
Probability and stochasticity

Today's agenda:

- Variability
- Probability theory
- Analyzing probability theory

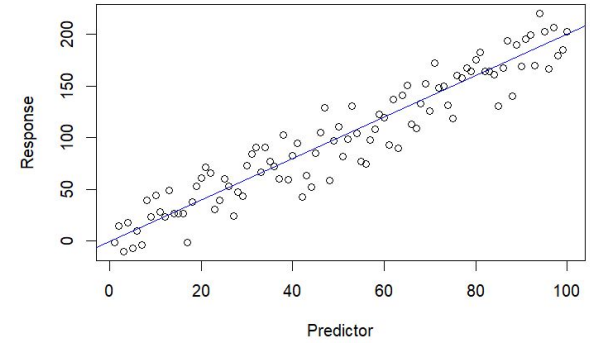
Variability

- Increasing recognition of importance biologically (e.g., intraspecific variation)
- Important component of statistical models (“signal” vs “noise”)



Variability

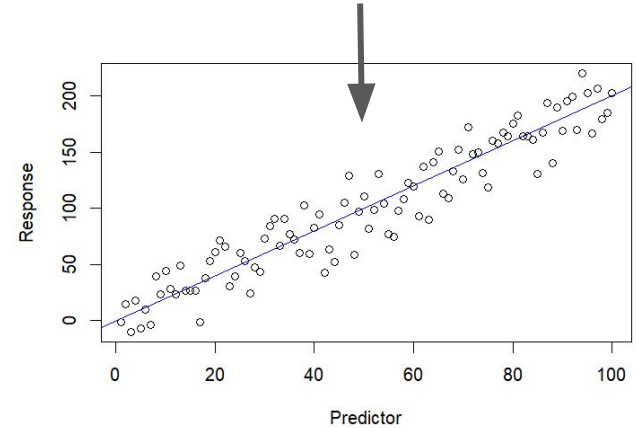
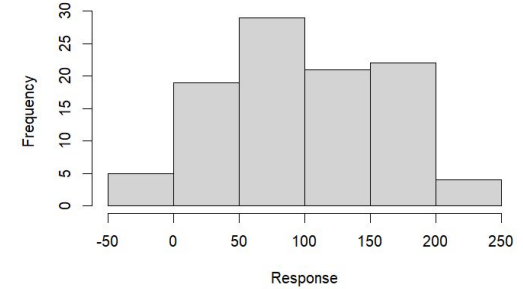
- Different sources of variability
 - Unmeasured variables (e.g., environmental stochasticity)
 - Measurement error
 - Inherent uncertainty
 - Demographic stochasticity (50% survival in a population of 3)
 - Nonlinear dynamics can increase/decrease variation



Variability

Different ways of handling variability

- Try to explain it vs. treat as error



Probability theory refresher

Probabilities sum to 1

Probability theory refresher

For mutually exclusive events: Probability A or B = $P(A) + P(B)$

$$P(\text{Dead or alive}) = P(\text{Dead}) + P(\text{Alive}) = 1$$

- If $P(\text{Dead}) = 0.4$, $P(\text{Alive}) = 0.6$

$$P(\text{SIR}) = P(S) + P(I) + P(R) = 1$$

- If $P(S) = 0.1$, and $P(R) = 0.2$, $P(S \text{ or } R) = 0.3$
- Then $P(I) = 0.7$

Probability theory refresher

For non-mutually exclusive events,

Have to account for double counting:

$P(\text{Blue OR Male}) =$

$$P(\text{Blue}) + P(\text{Male}) - P(\text{Blue AND Male}) = 0.65$$



Corrects for double-counts

	Male	Female
Blue	20	30
Green	15	35

Can you just add them?

$P(A \text{ or } B) = P(A) + P(B)$ only when A and B cannot both happen.

Which of these pairs can both happen at once?

1. A frog is infected / the frog is uninfected
2. A bird is male / the bird is blue
3. A site is a pond / the site is a stream
4. A tree flowered this year / the tree is more than 20 cm across

Probability theory refresher

Conditional probability: Probability of something, given something else

$$P(A|B) = P(A \text{ and } B) / P(B)$$

$$P(\text{Male}|\text{Blue}) = P(\text{Blue male}) / P(\text{Blue})$$

$$0.40 = 0.20/.50$$

	Male	Female
Blue	20	30
Green	15	35

Probability theory refresher

Independent variables: $P(A \text{ and } B) = P(A) * P(B)$

If Sex and Color were independent, we'd expect:

$$P(\text{Blue Male}) = P(\text{Blue}) * P(\text{Male})$$

$$0.175 = 0.5 * 0.35$$

0.20 $\neq$ 0.175, measurement error? dependence?

	Male	Female
Blue	20	30
Green	15	35 10

Practice: frogs and a fungus

100 frogs, each scored for habitat and for infection with *Bd*.

Working in groups, figure out:

1. $P(\text{infected})$
2. $P(\text{infected and pond})$
3. $P(\text{infected, given pond})$
4. $P(\text{infected, given stream})$
5. $P(\text{infected or pond})$

	Pond	Stream	Total
Infected	30	12	42
Uninfected	20	38	58
Total	50	50	100

Answers

1. $P(\text{infected}) = 42/100 = 0.42$

2. $P(\text{infected and pond}) = 30/100 = 0.30$

3. $P(\text{infected} \mid \text{pond}) = 0.30 / 0.50 = 0.60$

4. $P(\text{infected} \mid \text{stream}) = 0.12 / 0.50 = 0.24$

5. $P(\text{infected or pond}) = 0.42 + 0.50 - 0.30 = 0.62$

Pond frogs are infected at 60%, stream frogs at 24%.

P(A | B) is not P(B | A)

$$P(A|B) = P(A \text{ and } B) / P(B)$$

$$P(\text{infected} | \text{pond}) = 0.30 / 0.50 = 0.60$$

$$P(\text{pond} | \text{infected}) = 0.30 / 0.42 = 0.71$$

Same numerator. Different denominator. Different question.

P(spots | measles) is nearly 1.

P(measles | spots) is nearly 0.

Your turn: does habitat matter?

If infection and habitat were independent:

$$P(\text{infected and pond}) = P(\text{infected}) \times P(\text{pond})$$

- 1) Work out what that gives, and compare it to what we counted.
- 2) How many of the 100 frogs would be infected pond frogs?

	Pond	Stream	Total
Infected	30	12	42
Uninfected	20	38	58
Total	50	50	100

Answer: not independent

Expected if independent: $0.42 \times 0.50 = 0.21$, so 21 frogs

Actually counted: 30 frogs

More infected pond frogs than independence predicts.

Habitat and infection are associated.

Knowing the habitat changes the probability.

That is what a model is: $P(\text{outcome} \mid \text{what you know})$.

Questions so far?

What to know about probability distributions

Discrete:Poisson

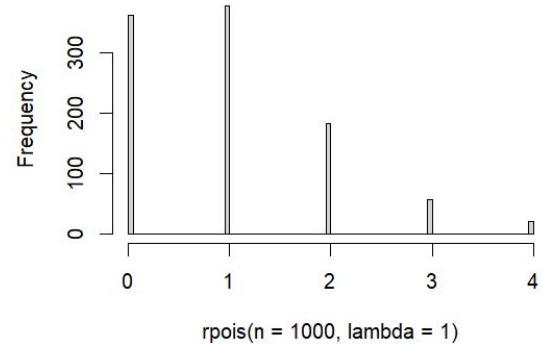
2 classes: Discrete vs continuous

Discrete:

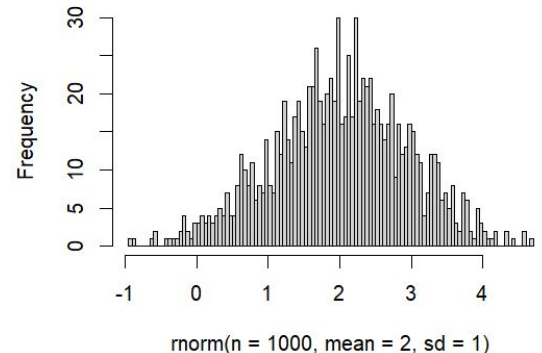
- **Integers**
- E.g., individuals, # events

Continuous:

- **Any real number** (or a subset, e.g. positives)
- E.g., body sizes, speeds, etc.
- $P(\text{any exact number}) = 0$
- Focus on $P(\text{some number or more extreme})$
 - E.g. $P(x \geq 4)$



Continuous:Normal



Continuous or discrete?

Organisms

- Body Mass
- Age
- Leaf Area
- Population size
- Population density
- Species richness
- Functional diversity
- Individuals per transects
- Mean body mass

Environment

- Temperature
- Elevation
- Depth
- Hurricane severity

Evolution

- Base pair substitution
- Evolutionary rate
- Species per taxon

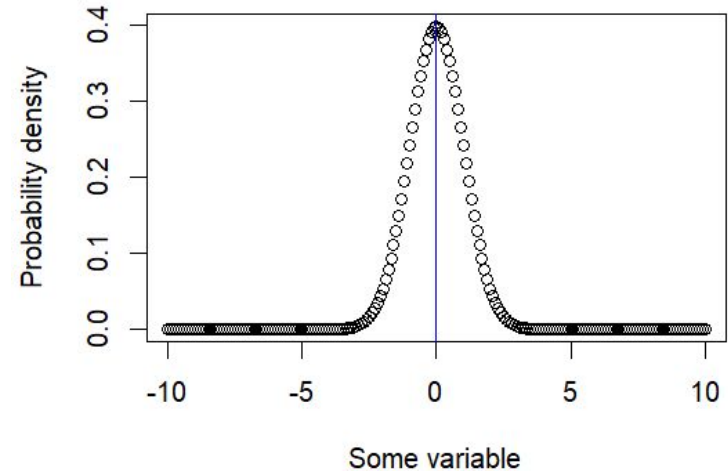
What to know about probability distributions

Central tendencies:

Mean

- “Average”
- Expected value (denoted $E[x]$)
- Also denoted $\bar{x}$

Use function `mean ( )`

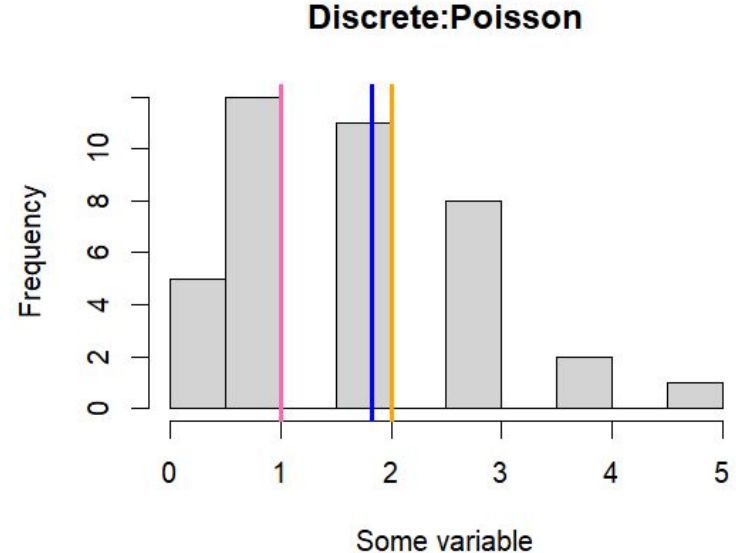


What to know about probability distributions

Central tendencies:

Median

- More robust than mean
- Less sensitive to outliers and skew
- Use function `median()`



Blue = mean

Orange = median

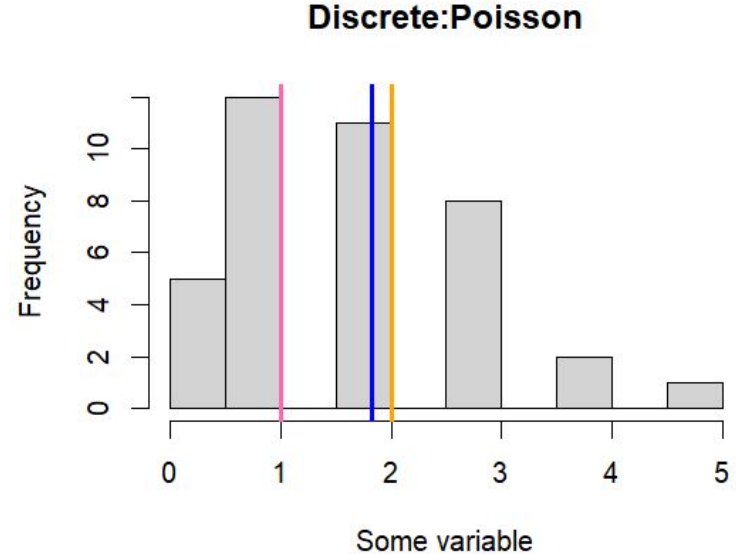
Pink = mode

What to know about probability distributions

Central tendencies:

Mode

- Most likely or most common value
- No built-in function
- DescTools package has `Mode()`



Blue = mean

Orange = median

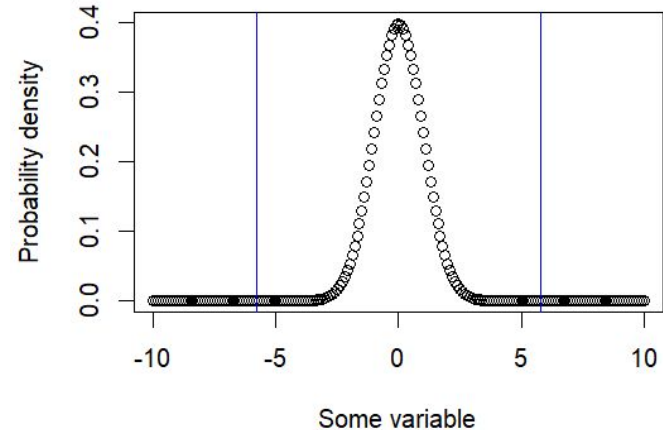
Pink = mode

What to know about probability distributions

Spread:

Variance and standard deviation

- Variance:
 - SD^2
 - Additive
 - Units of mean squared
 - Function `var()`
- SD
 - $\sqrt{\text{variance}}$
 - Not additive
 - Same units as mean
 - Function `sd()`



Blue lines: Mean +/- SD

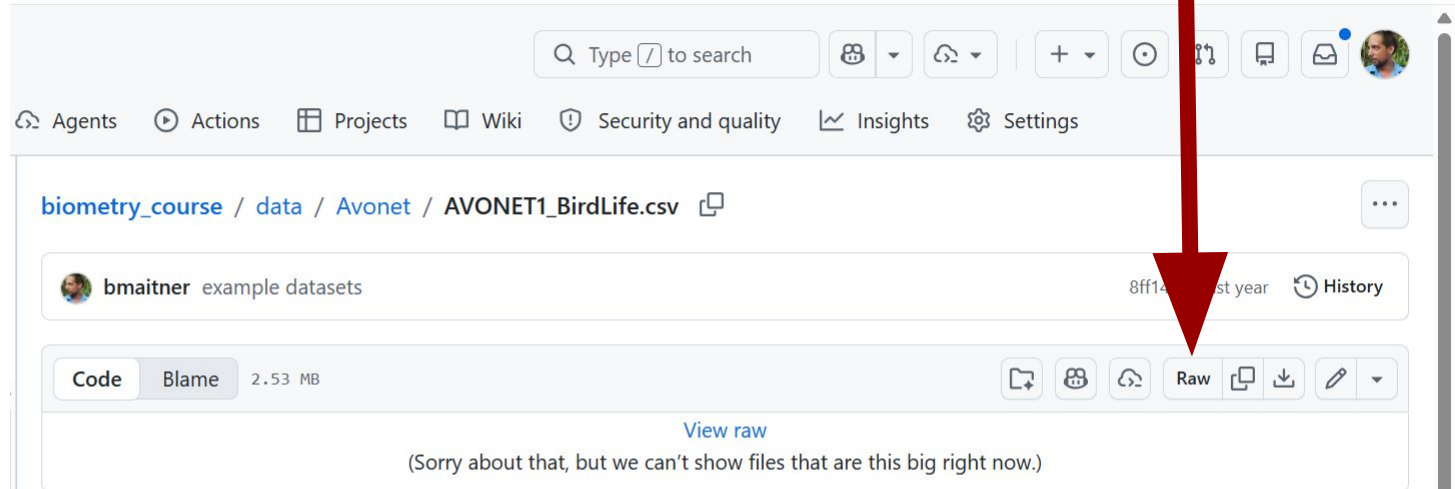
Activities

- Load data and calculate mean, variance, std dev, median, mode, etc.
 - Use data on github

Working with probability distributions

To get link:
Right click on “Raw”,
copy the URL

Load the avonet data



The screenshot shows a web-based file viewer interface. At the top, there is a search bar and navigation icons. Below this is a breadcrumb trail: [biometry_course](#) / [data](#) / [Avonet](#) / [AVONET1_BirdLife.csv](#). The file is owned by [bmaitner](#) and is part of the 'example datasets'. The file size is 2.53 MB. In the toolbar, the 'Raw' button is highlighted with a red arrow. Below the toolbar, there is a message: '(Sorry about that, but we can't show files that are this big right now.)'.

Working with probability distributions

Load the avonet data

```
avonet <-  
read.csv("https://github.com/bmaitner/biometry_course/raw/re  
fs/heads/main/data/Avonet/AVONET1_BirdLife.csv")
```

Or

```
library(readr)  
avonet <- read_rds("https://tinyurl.com/avonetbirddata")
```

Working with probability distributions

Using the avonet data:

- 1) Generate a histogram of Hand-Wing Index (high HWI = long-distance flyer)
 - a) Use `hist()` function
- 2) Calculate mean and median Hand-Wing Index
 - a) `mean()` and `median()`
- 3) Generate a histogram of Body Mass
 - a) Also try generating a histogram of the log of body mass (use `log10()` or `log()`)
- 4) Calculate mean and median body mass
 - a) How similar are they?
- 5) (optional) Try calculating the mode of body mass
 - a) `Mode()` in the DescTools package

Working with probability distributions

Using the avonet data:

- 1) Calculate the variance and standard deviation of Mass
 - a) `var()` and `sd()`
- 2) Check that variance equals SD^2
 - a) Use “==” to test whether things are equal
 - b) R uses “^” to denote raising something to a power, e.g., x^2 mean x^2
 - c) For square roots, use either `sqrt()` or $x^{0.5}$
- 3) Calculate the mean and variance of Range.Size

Remainder of class:

- Load in your focal data
- Use today's functions to better understand your key variables

Before next class:

- Skim 4.5 and 4.6, the bestiary of distributions